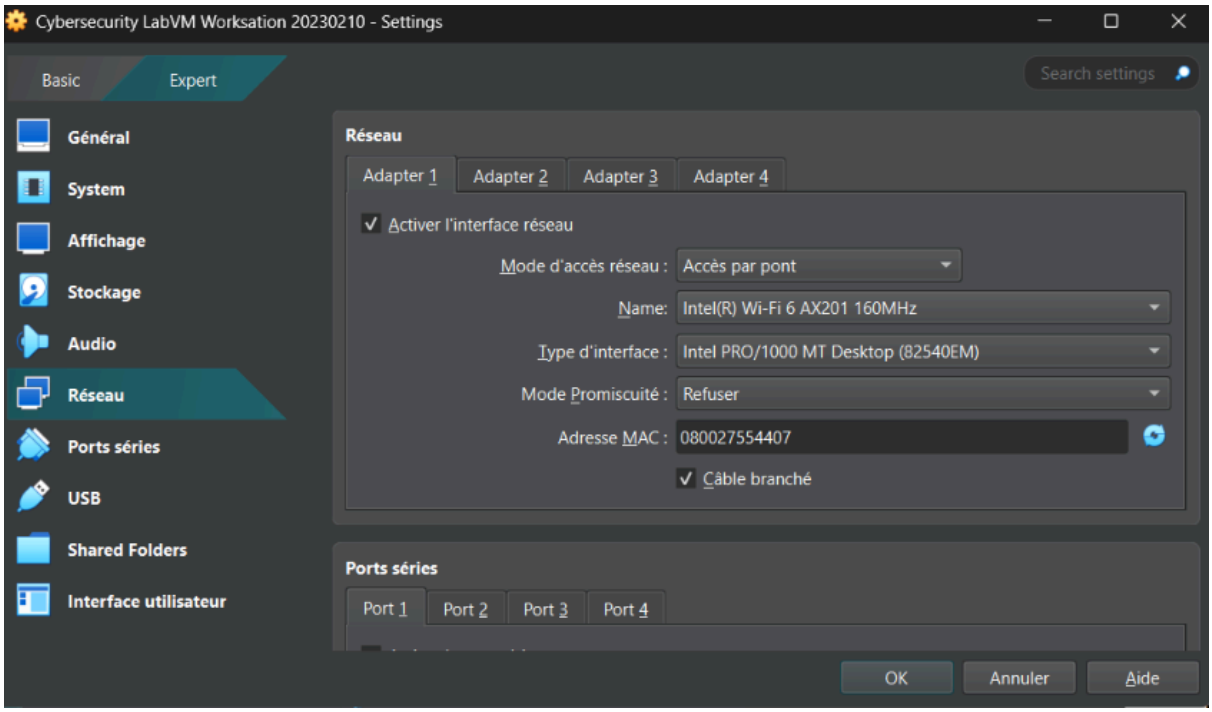
26.1.7-Lab - Snort and Firewall Rules

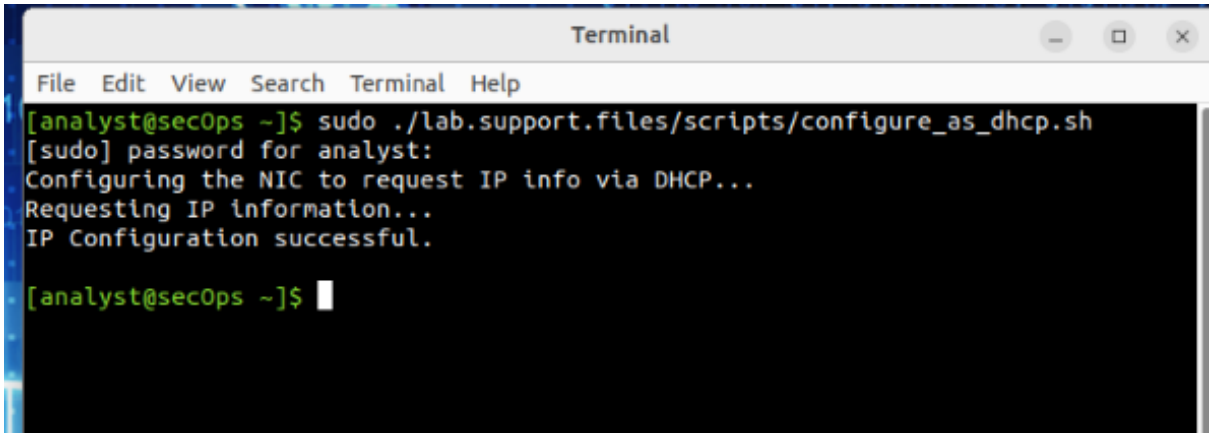
Ahmed Abdelhedi

Part 1: Preparing the Virtual Environment

a



b



```
[analyst@secOps ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.146  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::a00:27ff:fe55:4407  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:55:44:07  txqueuelen 1000  (Ethernet)
        RX packets 141  bytes 11105 (11.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 64  bytes 7679 (7.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 28  bytes 2239 (2.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 28  bytes 2239 (2.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

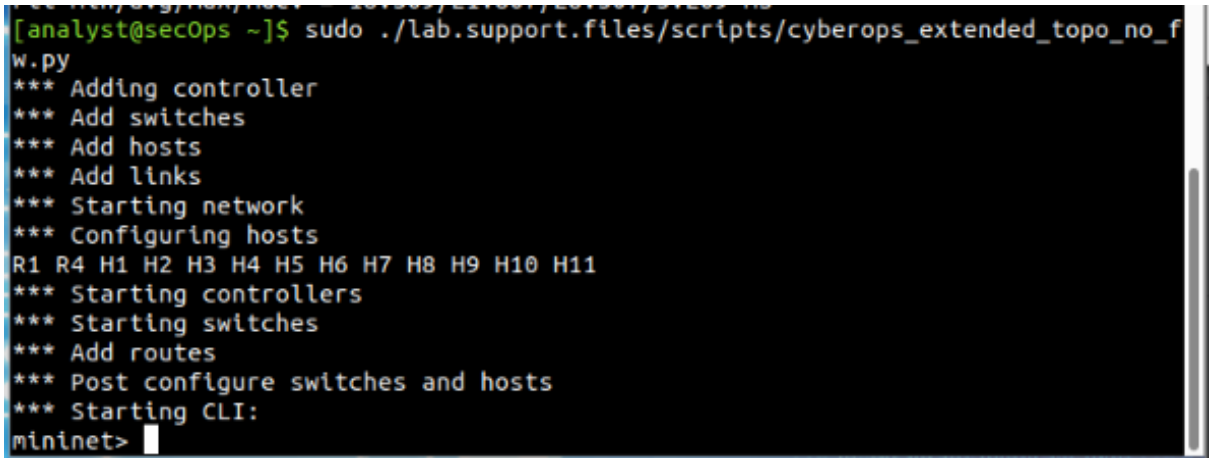
C

```
[analyst@secOps ~]$ ping www.cisco.com
PING e2867.dsca.akamaiedge.net (2.22.4.110) 56(84) bytes of data.
64 bytes from a2-22-4-110.deploy.static.akamaitechnologies.com (2.22.4.110): icm
p_seq=1 ttl=51 time=20.0 ms
64 bytes from a2-22-4-110.deploy.static.akamaitechnologies.com (2.22.4.110): icm
p_seq=2 ttl=51 time=28.3 ms
64 bytes from a2-22-4-110.deploy.static.akamaitechnologies.com (2.22.4.110): icm
p_seq=3 ttl=51 time=18.4 ms
64 bytes from a2-22-4-110.deploy.static.akamaitechnologies.com (2.22.4.110): icm
p_seq=4 ttl=51 time=21.2 ms
64 bytes from a2-22-4-110.deploy.static.akamaitechnologies.com (2.22.4.110): icm
p_seq=5 ttl=51 time=23.2 ms
64 bytes from a2-22-4-110.deploy.static.akamaitechnologies.com (2.22.4.110): icm
p_seq=6 ttl=51 time=19.7 ms
^C
--- e2867.dsca.akamaiedge.net ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5054ms
rtt min/avg/max/mdev = 18.369/21.807/28.307/3.269 ms
[analyst@secOps ~]$
```

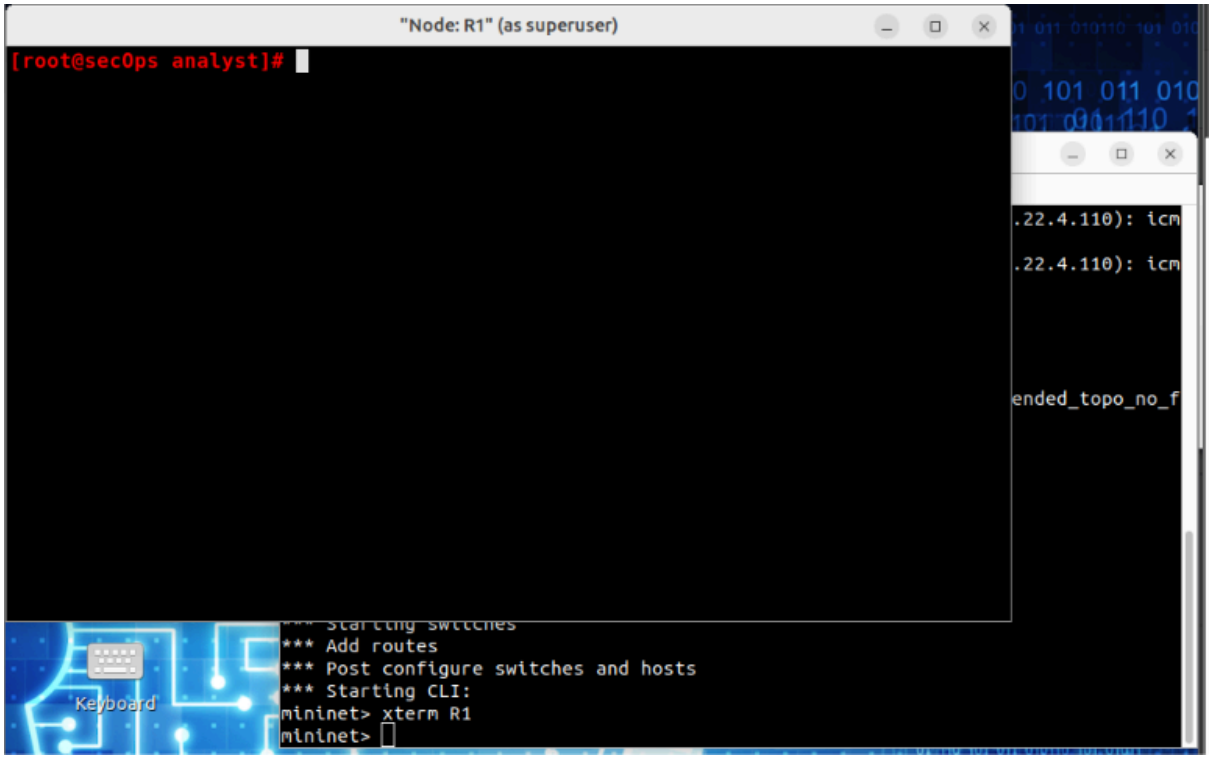
Part 2: Firewall and IDS Logs

Step 1: Real-Time IDS Log Monitoring

a

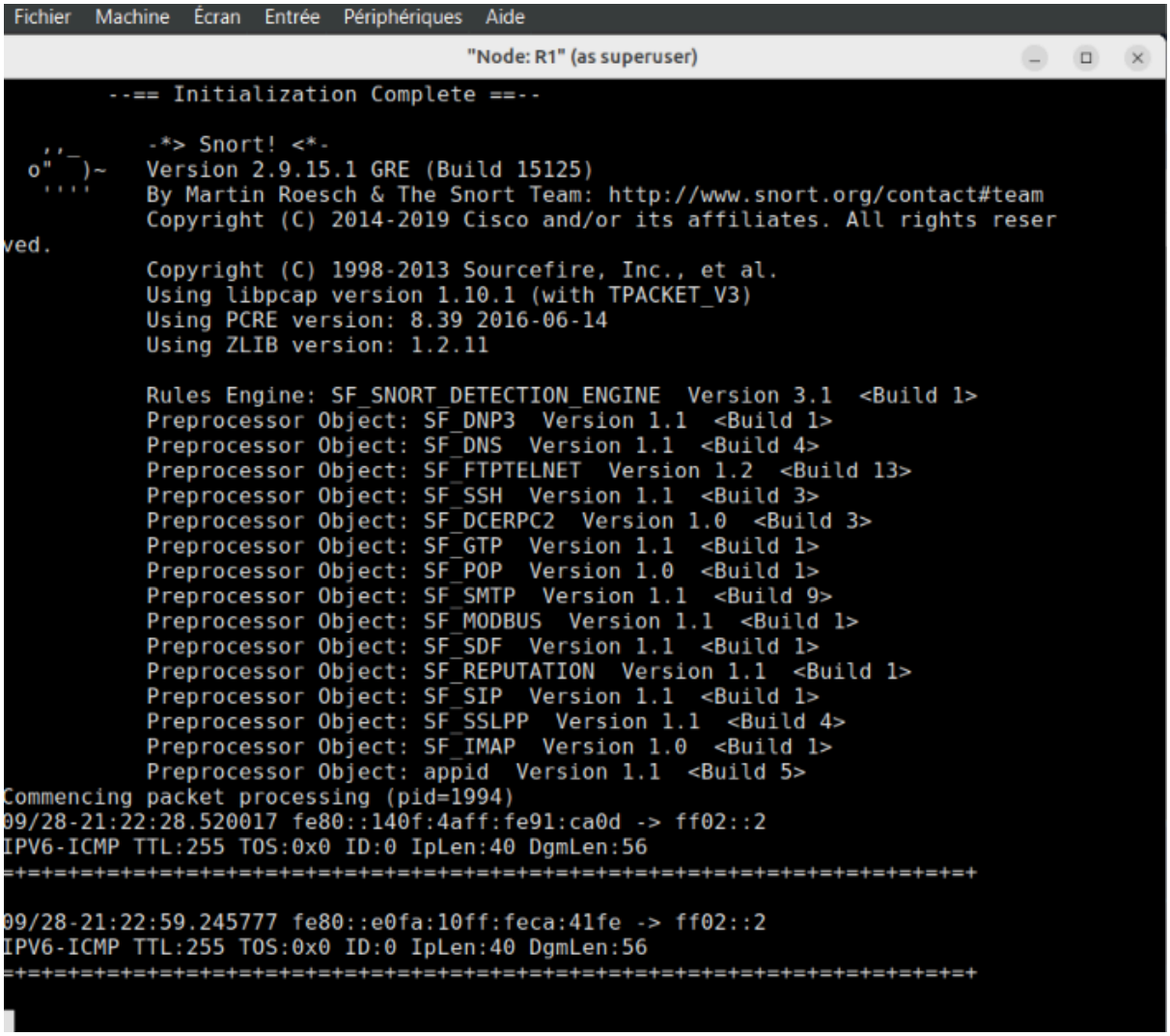


b

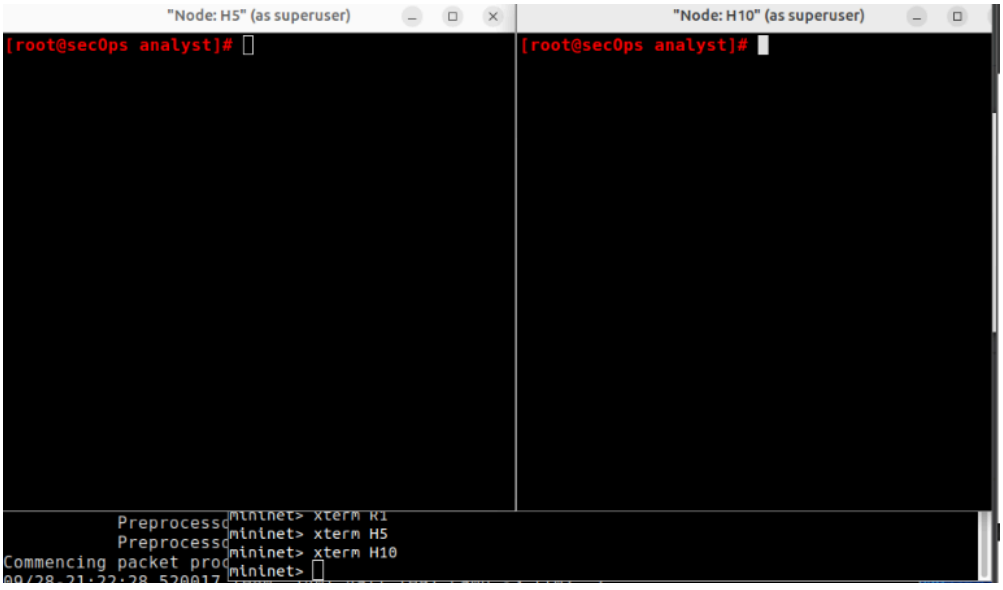


The user is logged in as **root**. The prompt [root@secOps analyst]# shows that the username is **root**, and the # symbol (instead of $) indicates **superuser privileges**

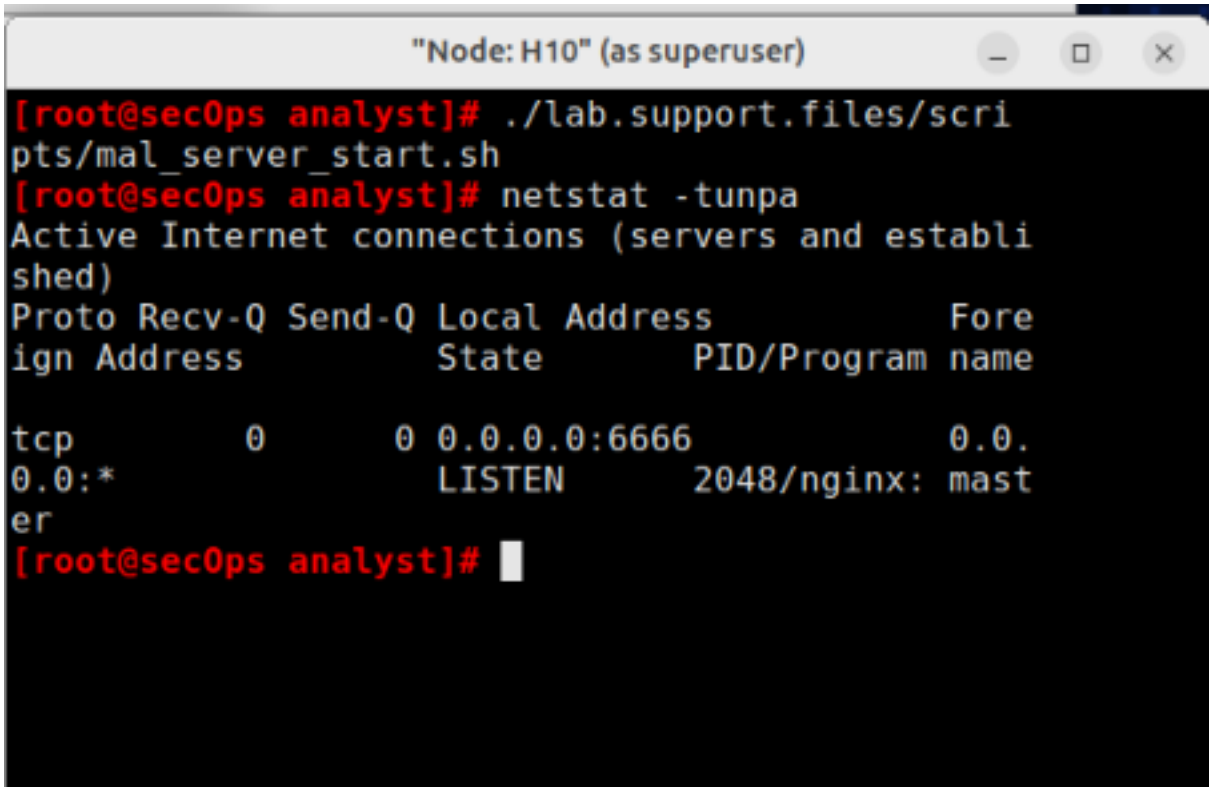
c


Fichier Machine Écran Entrée Périphériques Aide
"Node: R1" (as superuser)
--== Initialization Complete ==--
-*> Snort! <*-
Version 2.9.15.1 GRE (Build 15125)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reser
ved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11
Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.1 <Build 1>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: appid Version 1.1 <Build 5>
Commencing packet processing (pid=1994)
09/28-21:22:28.520017 fe80::140f:4aff:fe91:ca0d -> ff02::2
IPV6-ICMP TTL:255 TOS:0x0 ID:0 IpLen:40 DgmLen:56
09/28-21:22:59.245777 fe80::e0fa:10ff:feca:41fe -> ff02::2
IPV6-ICMP TTL:255 TOS:0x0 ID:0 IpLen:40 DgmLen:56

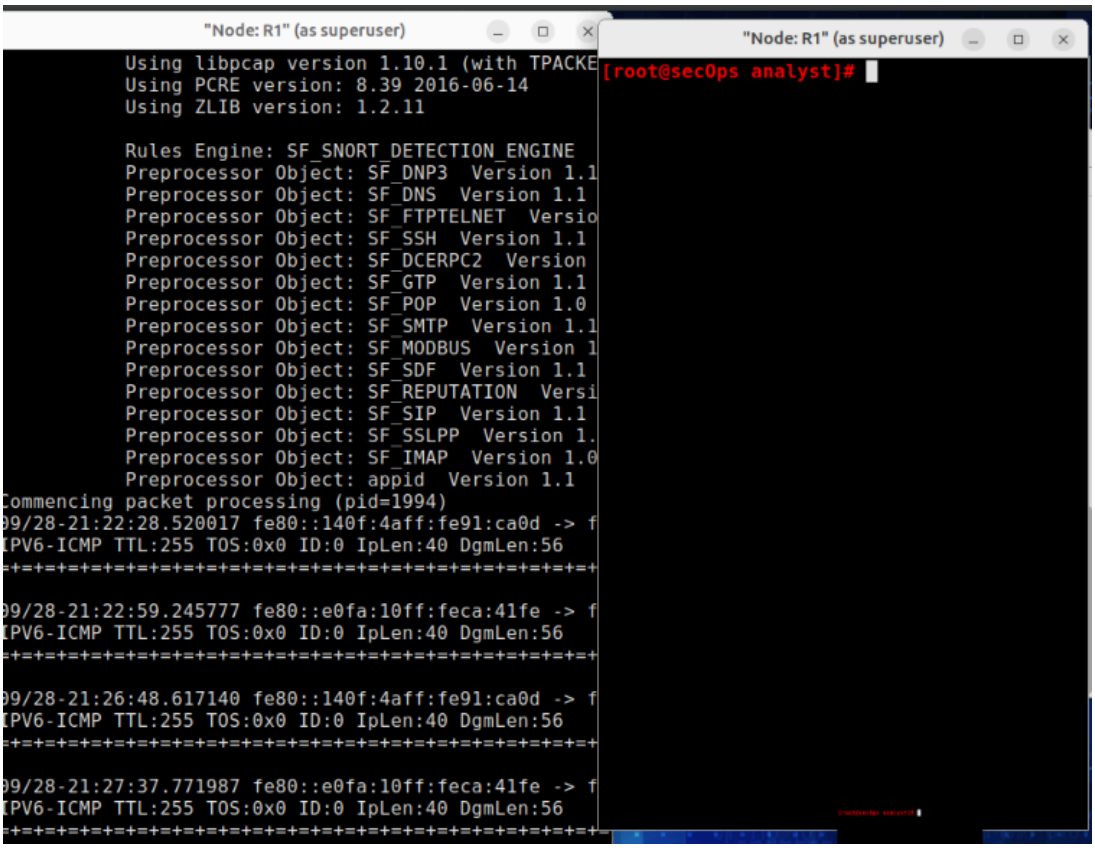
d


"Node: H5" (as superuser)
[root@secOps analyst]#
"Node: H10" (as superuser)
[root@secOps analyst]#
mininet> xterm R1
mininet> xterm H5
mininet> xterm H10
mininet>

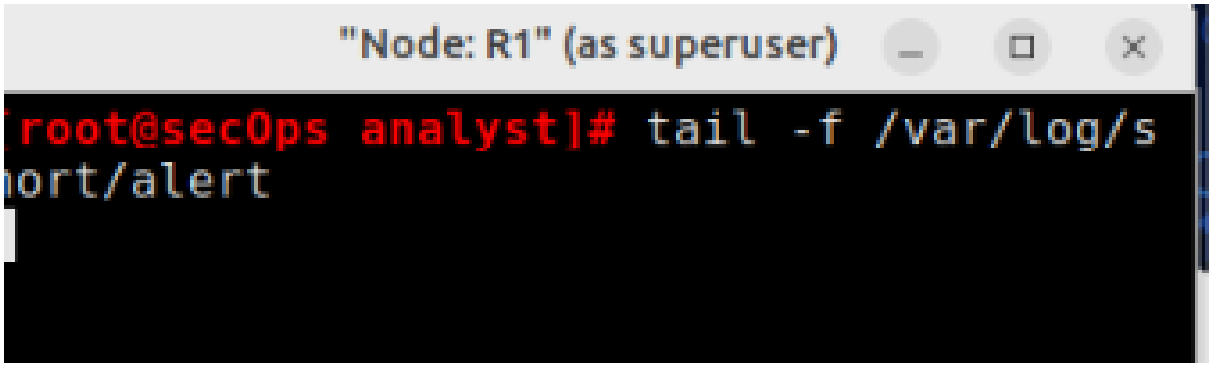
e+f


"Node: H10" (as superuser)
[root@secOps analyst]# ./lab.support.files/scripts/mal_server_start.sh
[root@secOps analyst]# netstat -tunpa
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address Foreign Address State PID/Program name
tcp 0 0 0.0.0.0:6666 0.0.0.0:* LISTEN 2048/nginx: master
[root@secOps analyst]#

g

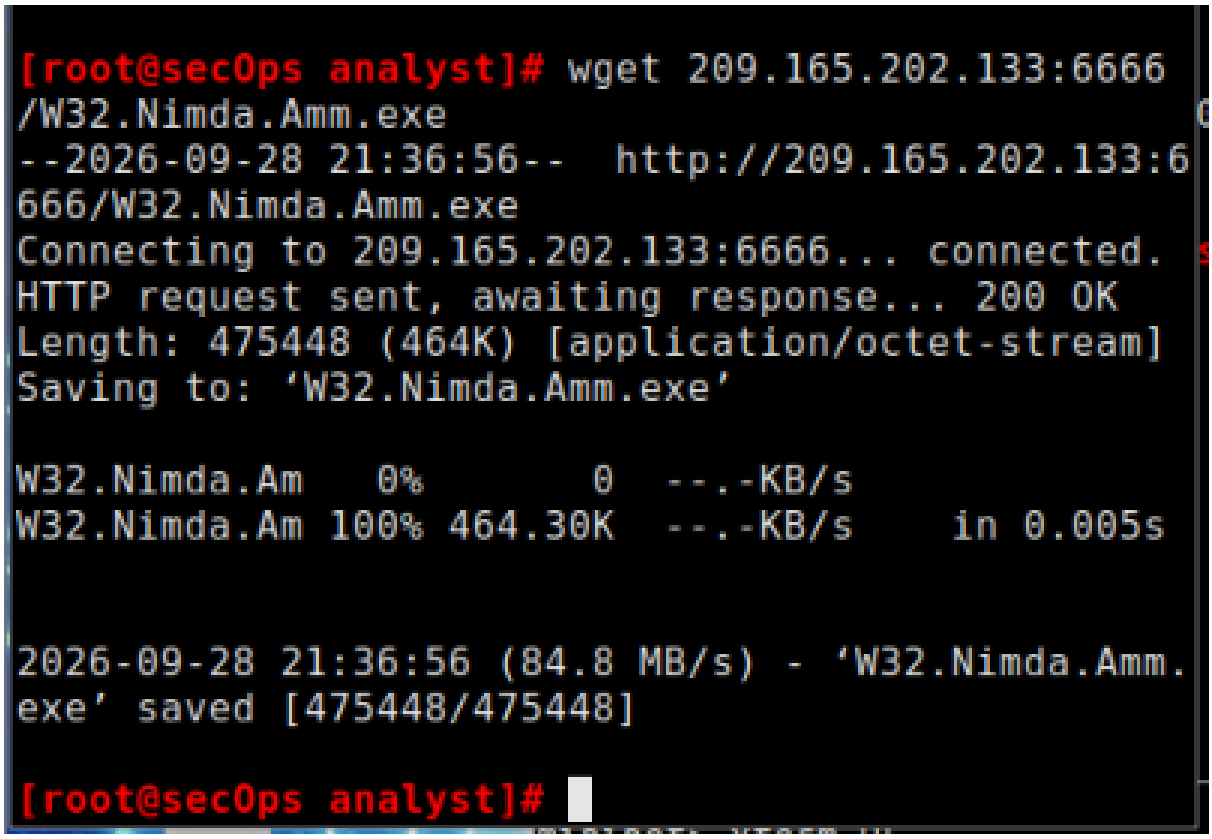

"Node: R1" (as superuser)
Using libpcap version 1.10.1 (with TPACKE
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11
Rules Engine: SF_SNORT_DETECTION_ENGINE
Preprocessor Object: SF_DNP3 Version 1.1
Preprocessor Object: SF_DNS Version 1.1
Preprocessor Object: SF_FTPTELNET Versio
Preprocessor Object: SF_SSH Version 1.1
Preprocessor Object: SF_DCERPC2 Version
Preprocessor Object: SF_GTP Version 1.1
Preprocessor Object: SF_POP Version 1.0
Preprocessor Object: SF_SMTP Version 1.1
Preprocessor Object: SF_MODBUS Version 1
Preprocessor Object: SF_SDF Version 1.1
Preprocessor Object: SF_REPUTATION Versi
Preprocessor Object: SF_SIP Version 1.1
Preprocessor Object: SF_SSLPP Version 1.
Preprocessor Object: SF_IMAP Version 1.0
Preprocessor Object: appid Version 1.1
ommencing packet processing (pid=1994)
9/28-21:22:28.520017 fe80::140f:4aff:fe91:ca0d -> f
IPV6-ICMP TTL:255 TOS:0x0 ID:0 IpLen:40 DgmLen:56
9/28-21:22:59.245777 fe80::e0fa:10ff:feca:41fe -> f
IPV6-ICMP TTL:255 TOS:0x0 ID:0 IpLen:40 DgmLen:56
9/28-21:26:48.617140 fe80::140f:4aff:fe91:ca0d -> f
IPV6-ICMP TTL:255 TOS:0x0 ID:0 IpLen:40 DgmLen:56
9/28-21:27:37.771987 fe80::e0fa:10ff:feca:41fe -> f
IPV6-ICMP TTL:255 TOS:0x0 ID:0 IpLen:40 DgmLen:56
"Node: R1" (as superuser)
[root@secOps analyst]#

h



no alerts !!!

i

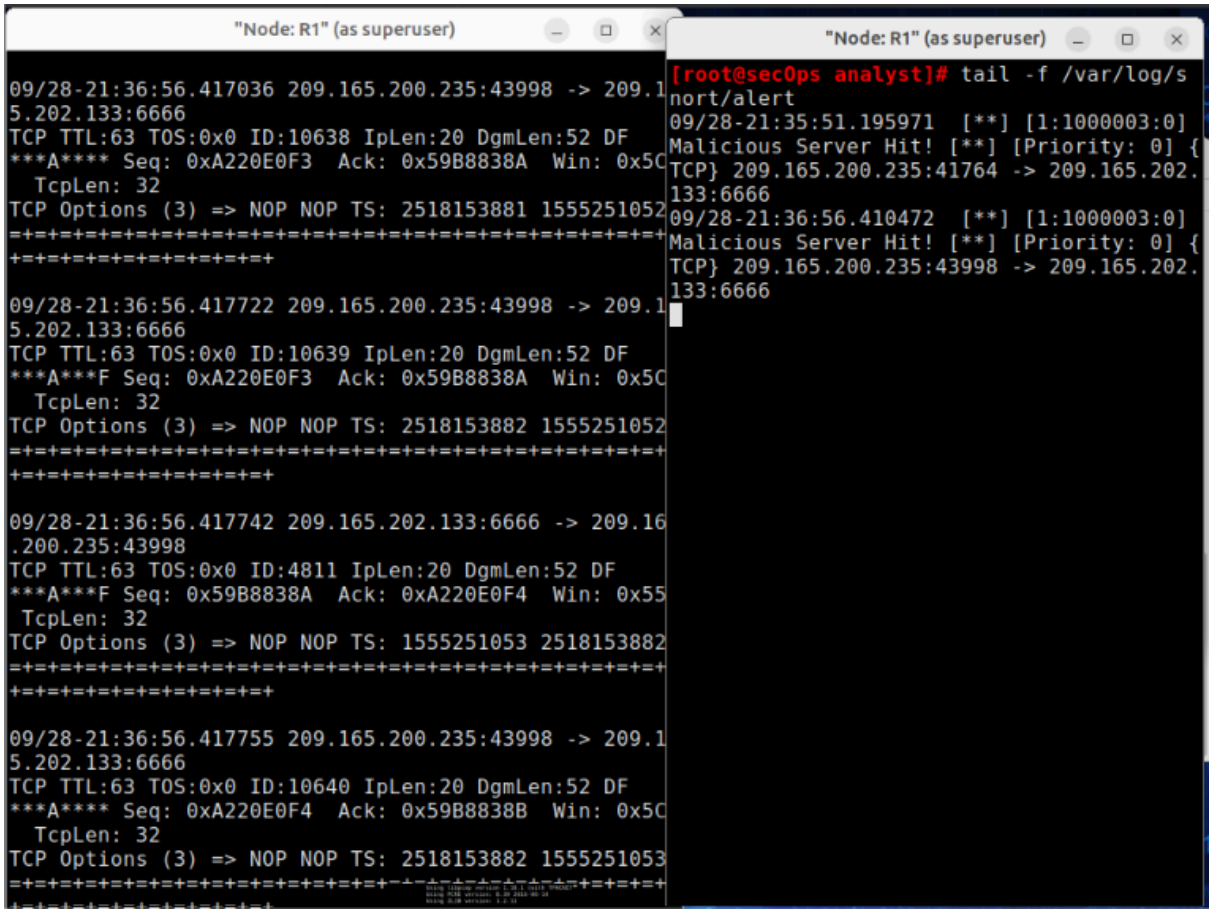


Port **TCP 6666**.

Yes. wget shows 200 OK, the progress bar reaches 100%, and the last line says saved [345088/345088], meaning all 345,088 bytes expected were received.

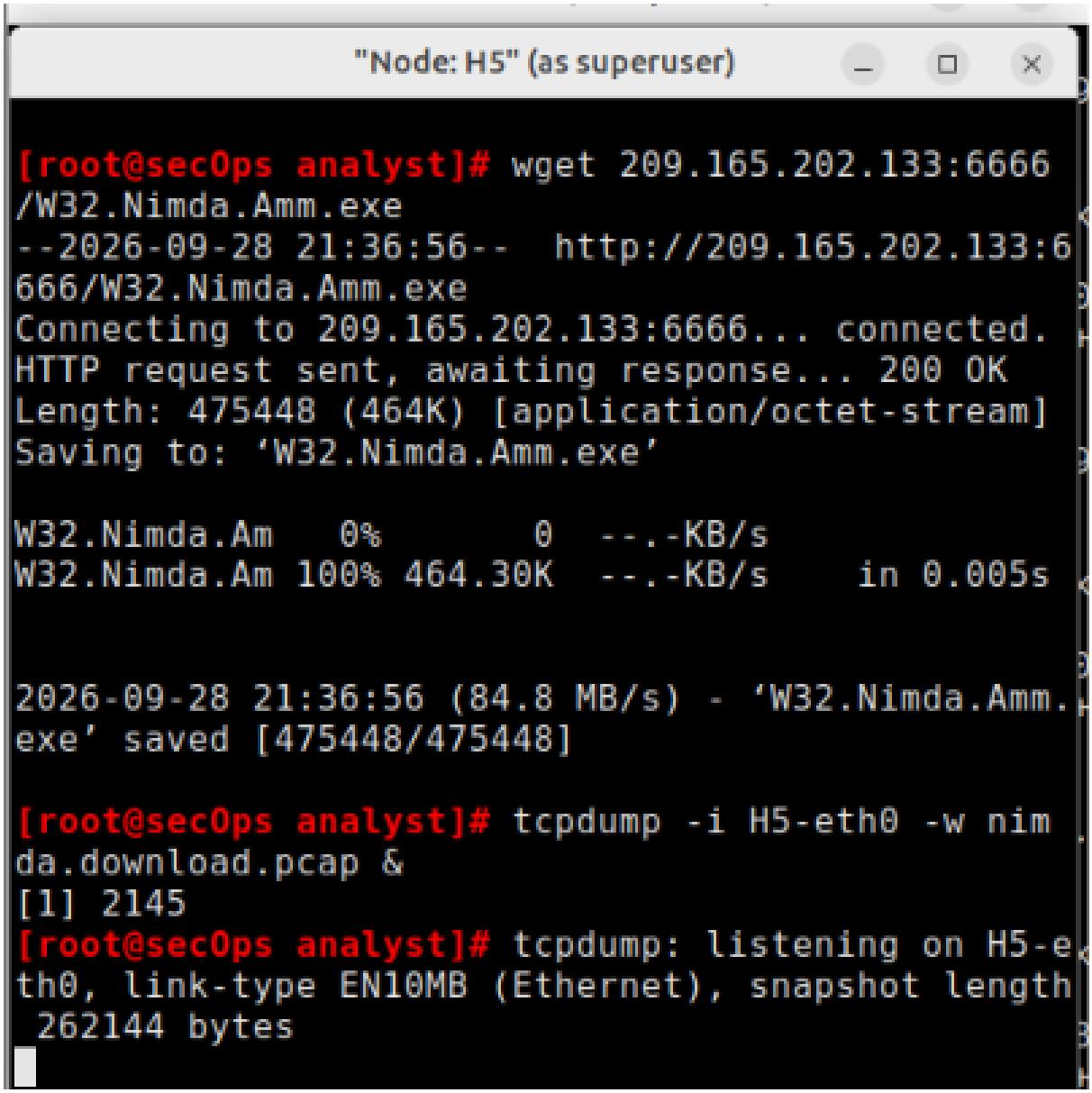
Yes. Snort inspected the payload as it passed through R1 and matched a signature, which wrote an alert to /var/log/snort/alert. It appears in the R1 window running tail -f: a "Malicious Server Hit!" alert (SID 1:1000003) from

209.165.200.235:xxxxx to 209.165.202.133:6666.

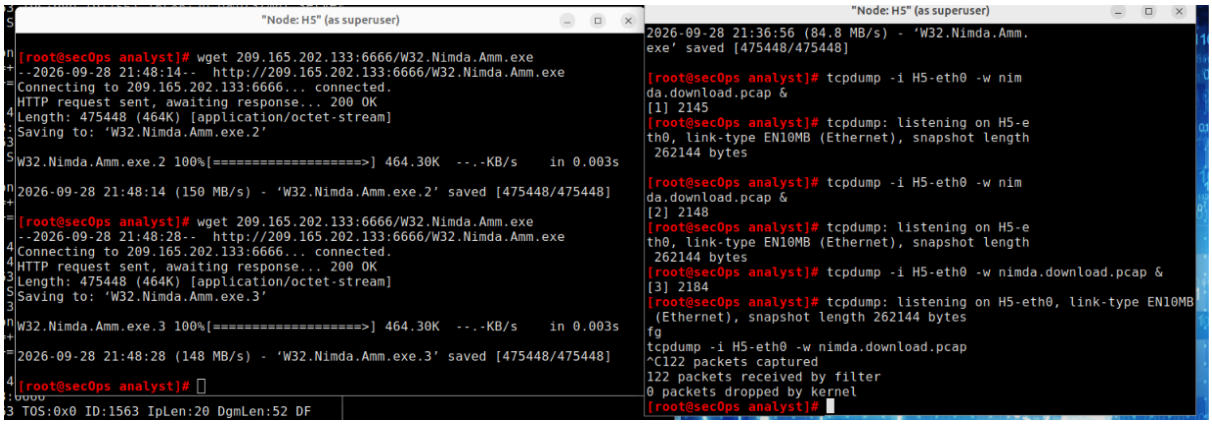


- **Source and destination IPv4 addresses:** source **209.165.200.235** (H5), destination **209.165.202.133** (H10).
- **Source and destination ports:** source **41764** (ephemeral), destination **6666**. For the second download, the source port is 43998.
- **When the download took place: September 28, at 21:35:51** (09/28-21:35:51.19571).
- **Message recorded by the signature: "Malicious Server Hit!"** (SID 1:1000003).

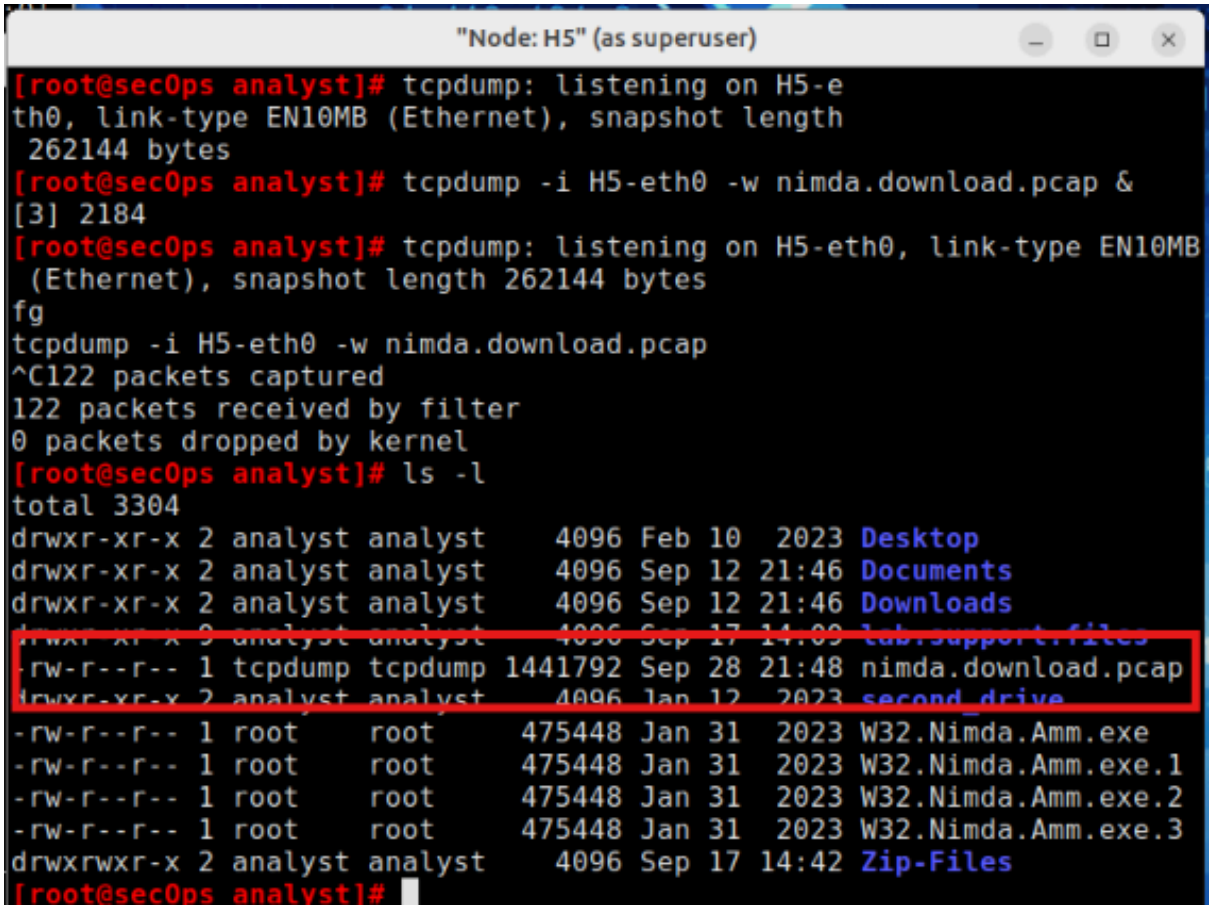
k+i


"Node: H5" (as superuser)
[root@secOps analyst]# wget 209.165.202.133:6666/W32.Nimda.Amm.exe
--2026-09-28 21:36:56-- http://209.165.202.133:6666/W32.Nimda.Amm.exe
Connecting to 209.165.202.133:6666... connected.
HTTP request sent, awaiting response... 200 OK
Length: 475448 (464K) [application/octet-stream]
Saving to: 'W32.Nimda.Amm.exe'
W32.Nimda.Am 0% 0 --.-KB/s
W32.Nimda.Am 100% 464.30K --.-KB/s in 0.005s
2026-09-28 21:36:56 (84.8 MB/s) - 'W32.Nimda.Amm.exe' saved [475448/475448]
[root@secOps analyst]# tcpdump -i H5-eth0 -w nimda.download.pcap &
[1] 2145
[root@secOps analyst]# tcpdump: listening on H5-eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes

m

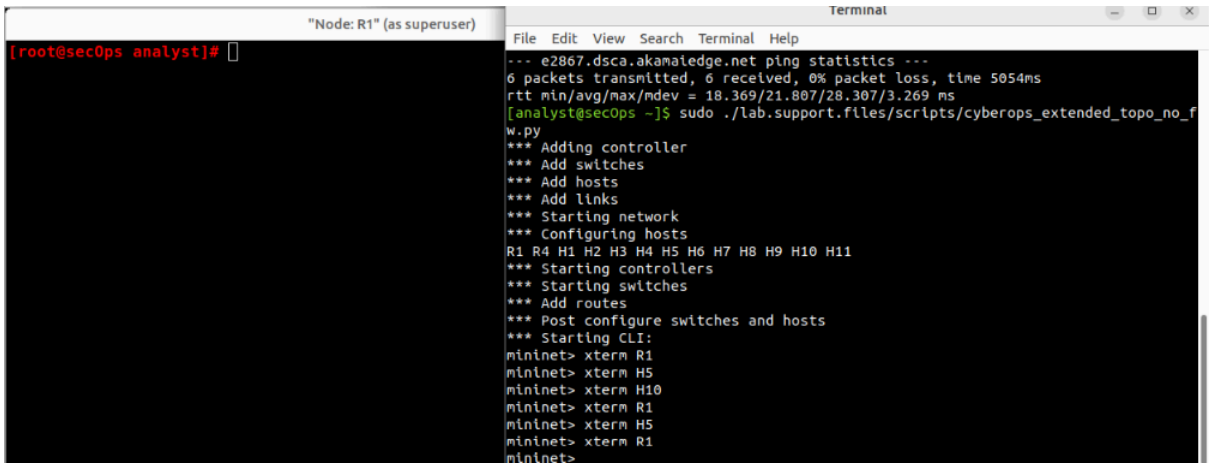

"Node: H5" (as superuser)
[root@secOps analyst]# wget 209.165.202.133:6666/W32.Nimda.Amm.exe
--2026-09-28 21:48:14-- http://209.165.202.133:6666/W32.Nimda.Amm.exe
Connecting to 209.165.202.133:6666... connected.
HTTP request sent, awaiting response... 200 OK
Length: 475448 (464K) [application/octet-stream]
Saving to: 'W32.Nimda.Amm.exe.2'
W32.Nimda.Amm.exe.2 100%[==========>] 464.30K --.-KB/s in 0.003s
2026-09-28 21:48:14 (150 MB/s) - 'W32.Nimda.Amm.exe.2' saved [475448/475448]
[root@secOps analyst]# wget 209.165.202.133:6666/W32.Nimda.Amm.exe
--2026-09-28 21:48:28-- http://209.165.202.133:6666/W32.Nimda.Amm.exe
Connecting to 209.165.202.133:6666... connected.
HTTP request sent, awaiting response... 200 OK
Length: 475448 (464K) [application/octet-stream]
Saving to: 'W32.Nimda.Amm.exe.3'
W32.Nimda.Amm.exe.3 100%[==========>] 464.30K --.-KB/s in 0.003s
2026-09-28 21:48:28 (148 MB/s) - 'W32.Nimda.Amm.exe.3' saved [475448/475448]
[root@secOps analyst]#
3 TOS:0x0 ID:1563 IpLen:20 DgmLen:52 DF
"Node: H5" (as superuser)
2026-09-28 21:36:56 (84.8 MB/s) - 'W32.Nimda.Amm.exe' saved [475448/475448]
[root@secOps analyst]# tcpdump -i H5-eth0 -w nimda.download.pcap &
[1] 2145
[root@secOps analyst]# tcpdump: listening on H5-eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
[root@secOps analyst]# tcpdump -i H5-eth0 -w nimda.download.pcap &
[2] 2148
[root@secOps analyst]# tcpdump: listening on H5-eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
[root@secOps analyst]# tcpdump -i H5-eth0 -w nimda.download.pcap &
[3] 2184
[root@secOps analyst]# tcpdump: listening on H5-eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
fg
tcpdump -i H5-eth0 -w nimda.download.pcap
^C122 packets captured
122 packets received by filter
0 packets dropped by kernel
[root@secOps analyst]#

n



Step 2: Tuning Firewall Rules Based on IDS Alerts

a



b

```
[root@secOps analyst]# iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

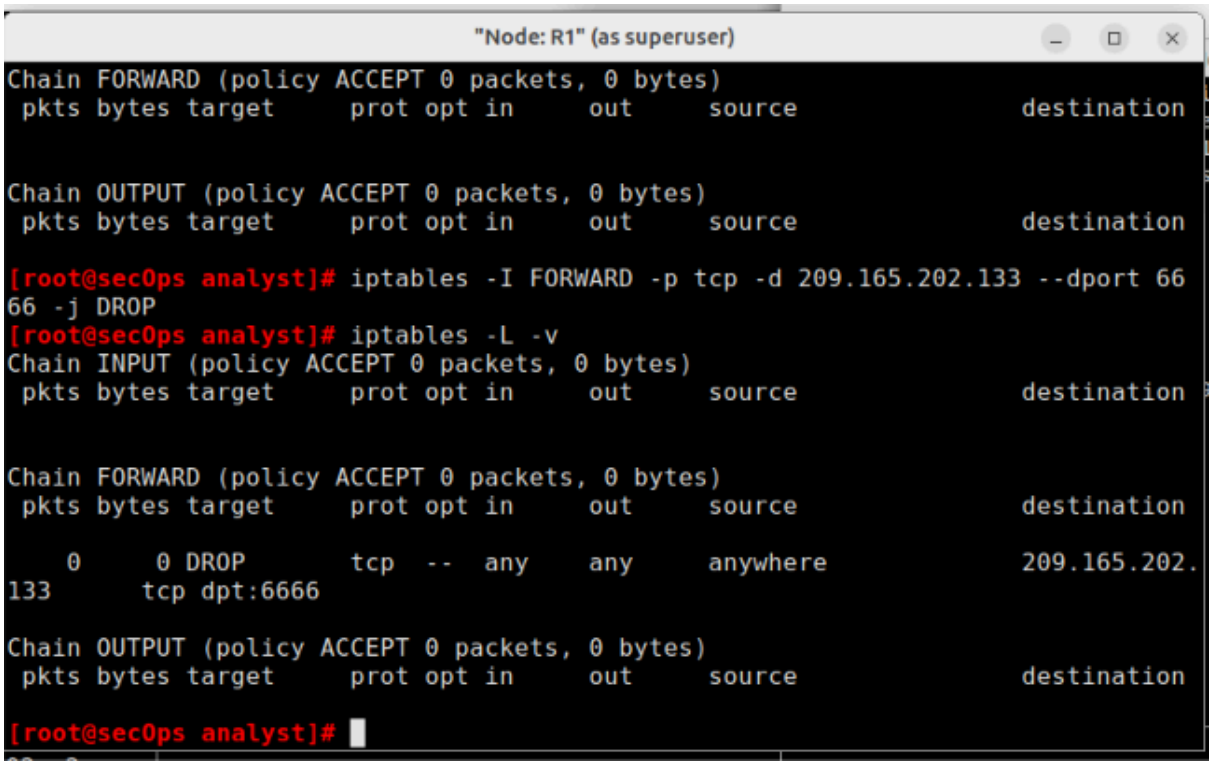
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

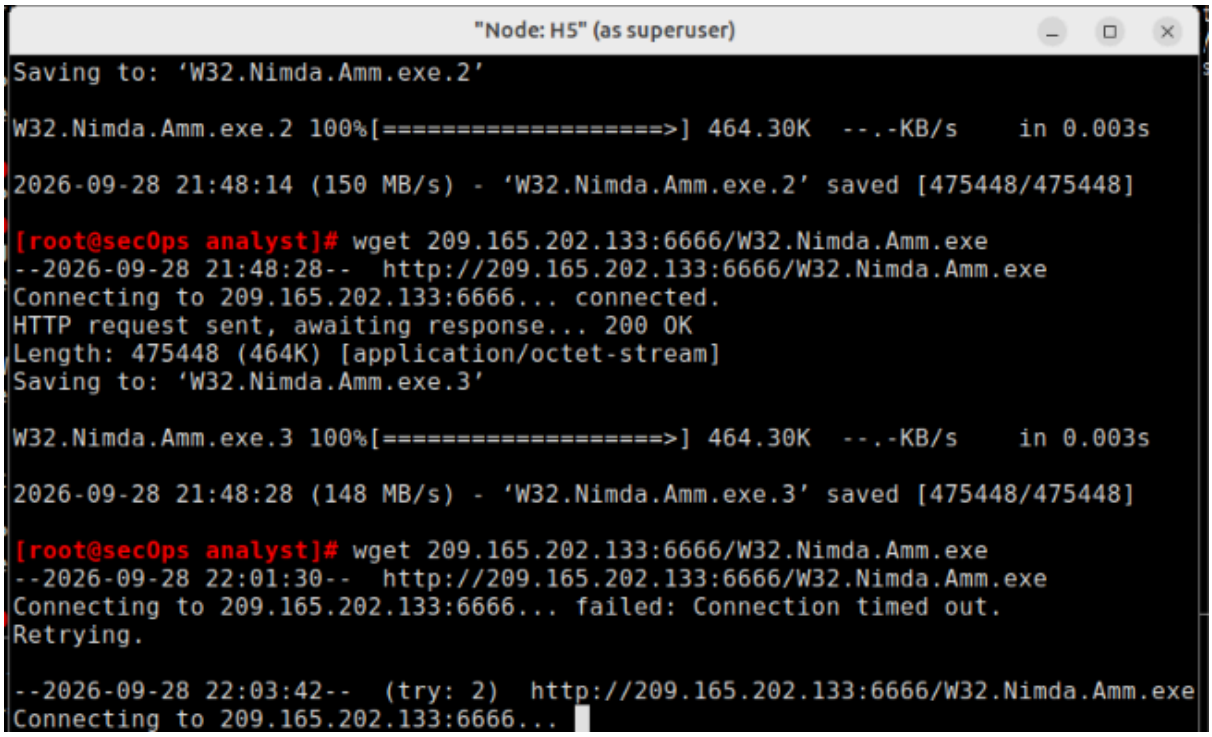
[root@secOps analyst]#
```

- **INPUT** (policy ACCEPT): traffic destined for R1 itself.
- **FORWARD** (policy ACCEPT): traffic passing through R1 to somewhere else.
- **OUTPUT** (policy ACCEPT): traffic originated by R1 itself.

c+d



e



Was the download successful this time? Explain:

no , because the download is deny by the new rule !

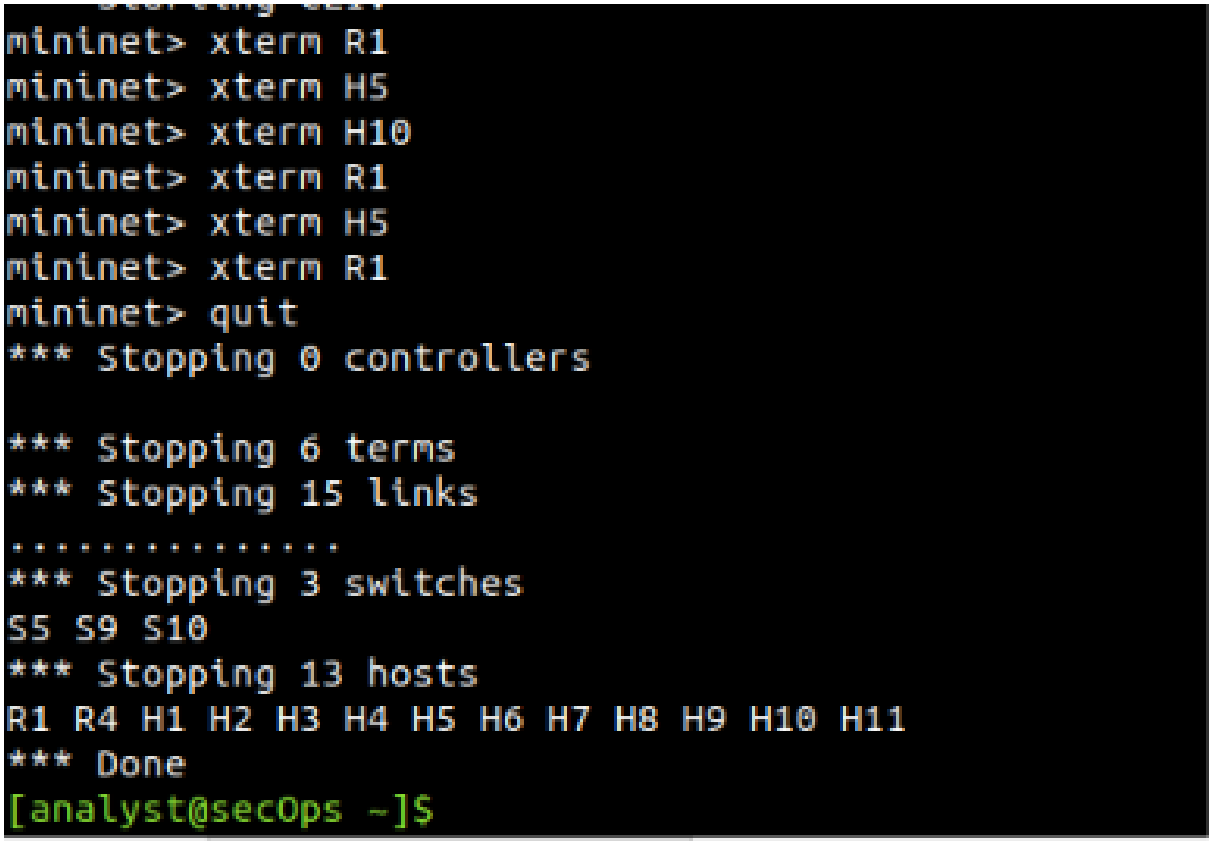
What would be a more aggressive but also valid approach when blocking the offending server?

Block **all traffic** to and from the server's IP address (209.165.202.133), regardless of protocol or port, instead of only TCP port 6666.

This is more aggressive because the attacker can no longer bypass the block by moving the malware to another port. An even broader option would be to block the whole subnet 209.165.202.128/27.

Part 3: Terminate and Clear Mininet Process

a



b

